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On universal metric spaces

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We present the following theorems (for the terms involved, see 1.1 and 1.2 below).

Theorem I. There is exactly one (up to isometry) complete ω -homogeneous ω -universal metric space of weight ω .

Theorem II. If $\alpha > \omega$ is a cardinal and $\alpha^\omega = \alpha$, then there exists exactly one (up to isometry) Urysohn universal space of weight α . Every Urysohn universal space of uncountable weight is complete.

Theorem III. If α is a cardinal, $\alpha^\omega > \alpha$, then there is no α -homogeneous α -universal space of weight α .

Theorem IV. There exists a meager Urysohn universal space of weight ω .

Theorem I has been proved by P. Urysohn, see [2] and [3]. In [3], P. Urysohn posed the question whether there exist non-complete ω -homogeneous strongly ω -universal spaces; this question is answered by Theorem IV. Theorem II can be easily deduced from Jónsson's theorems concerning relational systems (for the pertinent theorems and references to Jónsson's articles, see [1], §4).

Theorems II-IV are immediate consequences of propositions proved below: Theorem II follows from 2.8, 2.9, 2.7 and 2.10, Theorem III from 2.9 and 2.11, Theorem IV from 3.2.

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1.1. Notation and conventions. A) As a rule, the same symbol denotes a metric space and its underlying set, or a mapping $f: A \rightarrow B$ and the set $\{(a, b): a \in A, b = f(a)\}$, etc. - B) If $f: A \rightarrow B$ is a mapping, we often write fa instead of $f(a)$. - C) The class of all ordinals (all cardinals) will be denoted by Ord (respectively, by Card). The letters α, β and γ will stand for ordinals (for cardinals). The cardinality of a set A is denoted by $\text{card } A$. We have $\text{Card} \subset \text{Ord}$, and $\alpha = \{y: y < \alpha\}$ for all $\alpha \in \text{Ord}$. - D) If $\alpha \in \text{Ord}$, $\alpha \in \text{Card}$, we put $\alpha^\omega = \sup(\alpha^\beta: \beta < \alpha)$. - E) "Space" always means a metric space. The class of all non-void spaces is denoted by $\mathcal{M}$. The weight of a space S is denoted by wS . - F) If S and T are spaces, then $[S, T]$ denotes the set of all isometric mappings $f: S \rightarrow T$, and $\text{Is}(S, T)$ denotes the set of isometries $g: S \rightarrow T$, i.e. of bijective mappings $g \in [S, T]$. A mapping $h \in \text{Is}(S, S)$ will be called an isometry of S . If $\text{Is}(S, T) \neq \emptyset$, we write $S \cong T$.

1.2. Basic definitions. Let $\alpha \geq \omega$ be a cardinal. A space S will be called (1) α -homogeneous if, for any $f \in \text{Is}(A, B)$, where $A \subset S$, $B \subset S$, $\text{card } A < \alpha$, there is a $g \in \text{Is}(S, S)$ such that $g \upharpoonright A = f$, (2) α -universal (strongly α -universal) if $[T, S] \neq \emptyset$ whenever $T \in \mathcal{M}$ and $\text{card } T \leq \alpha$ (respectively, $wT \leq \alpha$). A space of weight α will be called Urysohn universal if it is α -homogeneous strongly α -universal.

1.3. Fact and notation. Let $(S, \rho) \in \mathcal{M}$. If $p \in S$, let $\hat{p}$ (or merely p) denote the function $s \mapsto \rho(p, s)$. Let $E'(S)$ denote the set of all $f: S \rightarrow \mathbb{R}$ satisfying $|fp - fq| \leq \rho(p, q) \leq fp + fq$ for all $p, q \in S$ and distinct from all $\hat{s}, s \in S$. Put $E(S) = S \cup E'(S)$. If $p, q \in E(S)$, put $\rho^E(p, q) = \sup\{|p(s) - q(s)|: s \in S\}$. Then ρ^E is a metric on $E(S)$. The space $(E(S), \rho^E)$ will be denoted by $E(S, \rho)$ or by $E(S)$.

1.4. Fact and notation. Let $S \in \mathcal{M}$, $\emptyset \neq T \subset S$. If $f \in E'(T)$, let f^* denote the function $s \mapsto \inf\{\rho(s, t) + ft: t \in T\}$ defined on S . If $f^* = \hat{p}$ for some $p \in S$, put $f_S = p$; if not, put $f_S = f^*$. If $t \in T$, put $t_S = t$. Then the mapping $r \mapsto r_S$, defined on $E(T)$, is in $[E(T), E(S)]$.

Proof. It is easy to see that either $f^* \in E'(S)$ or $f^* = \hat{p}$ for some $p \in S$. We are going to show that if $f, g \in E'(T)$ and $f^*, g^* \in$

$\in E'(S)$, then $\sup\{|f^*s - g^*s|: s \in S\} \leq \rho^E(f, g)$; the remaining assertions are easy to prove. Suppose there are $s \in S$ and $\varepsilon > 0$ such that $|f^*s - g^*s| > \rho^E(f, g) + \varepsilon$. We can assume $f^*s > g^*s$. Choose $t \in T$ such that $gt + \rho(s, t) < g^*s + \varepsilon$. Since $ft + \rho(s, t) \geq f^*s$, we get $ft - gt \geq f^*s + g^*s - \varepsilon > \rho^E(f, g)$, which is a contradiction.

1.5. Notation. Let $S \in \mathcal{M}$ and let $\alpha > 1$. The subspace of $E(S)$ consisting of all t_S , where $t \in \bigcup\{E(T): T \subset S, 0 < \text{card } T < \alpha\}$, will be denoted by $E(S, \alpha)$.

1.6. Fact and notation. Let $S \in \mathcal{M}$ and let $\varphi \in \text{Is}(S, S)$. Then there is exactly one $\Phi \in \text{Is}(E(S), E(S))$ such that $\Phi \upharpoonright S = \varphi$. The isometry Φ will be denoted by $E(\varphi)$. For any $\alpha > 1$, $E(S, \alpha)$ is invariant under Φ .

Proof. If $s \in S$, put $\Phi(s) = \varphi s$. If $f \in E'(S)$, put $\Phi(f) = f \circ \varphi^{-1}$. It is easy to see that Φ is an isometry of $E(S)$. - Let Ψ be an isometry of $E(S)$ and let $\Psi \upharpoonright S = \varphi$. Let $f \in E'(S)$, $g = \Psi(f)$. It is easy to see that, for any $h \in E'(S)$ and any $p \in S$, $\rho^E(h, p) = h(p)$. For any $s \in S$, we have $\rho^E(g, \varphi s) = \rho^E(f, s)$, hence $g(\varphi s) = f(s)$, $(g \circ \varphi)(s) = f(s)$. This implies $g \circ \varphi = f$, $g = f \circ \varphi^{-1}$.

1.7. Fact. Let $(S, \rho) \in \mathcal{M}$, $A \subset S$, $0 < \text{card } A < \omega$, $B \subset S$. Let $\varphi: A \rightarrow B$ be a bijection. Let $\varepsilon > 0$ and let $\rho(a, \varphi a) \leq \varepsilon$ for each $a \in A$. If $f \in E'(A)$, $g \in E'(B)$ and $f = g \circ \varphi$, then $\rho^E(f_S, g_S) \leq \varepsilon$.

Proof. Let $s \in S$. We have $f_S(s) = \inf\{ft + \rho(s, t): t \in A\}$, $g_S(s) = \inf\{g(\varphi t) + \rho(s, \varphi t): t \in A\} = \inf\{ft + \rho(s, \varphi t): t \in A\}$. Since $|\rho(s, t) - \rho(s, \varphi t)| \leq \rho(t, \varphi t) \leq \varepsilon$ for each $t \in A$, we get $|f_S(s) - g_S(s)| \leq \varepsilon$.

1.8. Lemma. Let $1 < \alpha \leq \omega$. Let $S \in \mathcal{M}$. Then $w(E(S, \alpha)) = \omega \cdot wS$, $\text{card } E(S, \alpha) = 2^\omega \cdot \text{card } S$.

Proof. Since $E(S, \omega) = \bigcup\{E(S, n): 1 < n < \omega\}$, we can assume that $\alpha = n < \omega$. If $A \subset S$, $0 < \text{card } A < n$, put $P(A) = \{u_S: u \in E(A)\}$. Since $w(E(A)) = \omega$, $\text{card } E(A) = 2^\omega$, we get, by 1.4, $w(P(A)) = \omega$, $\text{card } P(A) = 2^\omega$. Let $T \subset S$, $\text{card } T = wS$. By 1.7, the union of $P(A)$, where $A \subset T$, $0 < \text{card } A < n$, is dense in $E(S, n)$. This proves $wE(S, n) \leq \omega \cdot wS$; evidently, $wE(S, n) \geq \omega \cdot wS$. Finally, we have $E(S, n) = \bigcup\{E(T): T \subset S, 0 < \text{card } T < n\}$, hence $\text{card } E(S, n) \leq 2^\omega \cdot \text{card } S$.

1.9. Fact and notation. Let $S \in \mathcal{M}$. There exists exactly one indexed class $(S_x: x \in \text{Ord})$ such that (1) $S_0 = S$, (2) if $x \in \text{Ord}$, then $S_{x+1} = E(S_x, \omega \cdot (\text{card } x)^+)$, (3) if x is a limit ordinal, then $S_x = \bigcup (S_y: y < x)$. - The spaces S_x will be denoted by $U_x(S)$.

1.10. Lemma. Let $S \in \mathcal{M}$, $\text{card } S \leq 2^\omega$. Then, for any ordinal $x > \omega$, $\text{card } U_x(S) \leq 2^x$.

Proof. By 1.8, $\text{card } U_\omega(S) = 2^\omega$. Put $V = \{x: x > \omega, \text{card } U_x(S) > 2^x\}$. Suppose $V \neq \emptyset$ and put $y = \min V$. Put $\beta = \text{card } y$. If y is a limit ordinal, then $U_y(S) = \bigcup (U_x(S): x < y)$, hence $\text{card } U_y(S) \leq \beta \cdot 2^y$, which contradicts $y \in V$. If $y = z + 1$, then $U_y(S) = E(U_z(S), \beta^+)$, hence $\text{card } U_y(S) = \alpha^\beta$, where $\alpha = \text{card } U_z(S)$. If $z \in \text{Card}$, then, clearly, $\alpha^\beta = \alpha^z = 2^z = 2^y$. If $z \notin \text{Card}$, then $\alpha = 2^\beta$, $\alpha^\beta = 2^\beta = 2^y$. Hence $\text{card } U_y(S) \leq 2^y$, which contradicts the supposition.

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2.1. Notation. If α and β are cardinals, then $\mathcal{E}(\alpha, \beta)$ will denote the class of all $(T, S) \in \mathcal{M} \times \mathcal{M}$ such that $T \subset S$ and if $M \in \mathcal{M}$, $A \subset M$, $\text{card } A < \alpha$, $\text{card } (M \setminus A) < \beta$, $f \in [A, T]$, then $g \upharpoonright A = f$ for some $g \in [M, S]$.

2.2. Lemma. Let $\alpha \geq \omega$ be a cardinal. Let $S \in \mathcal{M}$. Let S_x , $x < \alpha$, be subspaces of S . Assume that (1) $\bigcup S_x = S$, (2) if $x < \alpha$, then $(S_x, S_y) \in \mathcal{E}(\alpha, 2)$ for some $y < \alpha$, (3) if $K \subset S$, $\text{card } K < \alpha$, then $K \subset S_z$ for some $z < \alpha$. Then $(S, S) \in \mathcal{E}(\alpha, \alpha^+)$.

Proof. Put $\beta = \alpha^+$. Let M, A and f be as in 2.1. Put $\gamma = \text{card } (M \setminus A)$. Choose a bijection $h: \gamma \rightarrow M \setminus A$. For $x \leq \gamma$ let G_x consist of all $g \in [A \cup h\{y: y < x\}, S]$ such that $g \upharpoonright A = f$. Put $G = \bigcup G_x$. It is easy to see that G , ordered by inclusion (cf. 1.1 A), contains a maximal element, say k . Suppose that $k \in G_x$, where $x < \gamma$. Then $\text{dom } k = A \cup h\{y: y < x\}$, hence $\text{card } (\text{dom } k) < \alpha$ and therefore $\text{dom } k \subset S_y$ for some $y < \alpha$. Choose a $z < \alpha$ such that $(S_y, S_z) \in \mathcal{E}(\alpha, 2)$. Then there exists a $k^* \in [(\text{dom } k) \cup \{hx\}, S]$ such that $k^* \upharpoonright \text{dom } g = g$. This is a contradiction, for $k^* \in G_{x+1} \subset G$. We have shown that $k \in G_\gamma$, which proves the lemma.

2.3. Corollary. For any cardinal $\alpha \geq \omega$, $\mathcal{E}(\alpha, 2) = \mathcal{E}(\alpha, \alpha^+)$.

2.4. Notation. If $\alpha \geq \omega$, we put $\mathcal{E}(\alpha) = \{(S, S) \in \mathcal{E}(\alpha, 2)\}$.

2.5. Lemma. Let $\alpha \geq \omega$. Let $S_i \in \mathcal{E}(\alpha)$, $i = 1, 2$. Let $A_i \subset S_i$, $\text{card } A_i < \alpha$ and let $f \in \text{Is}(A_1, A_2)$. Let $K_i \subset S_i$, $\text{card } K_i \leq \alpha$. Then there exist subspaces $T_i \subset S_i$, and an isometry $g: T_1 \rightarrow T_2$ such that $A_i \cup K_i \subset T_i$ and $g \upharpoonright A_1 = f$.

Proof. Clearly, we can assume that $K_i \cap A_i = \emptyset$, $\text{card } K_i = \text{card } K_2 \geq \omega$. Put $\beta = \text{card } K_1$ and choose bijections $u: \{x: x < \beta\} \rightarrow K_1$, $v: \{x: x < \beta\} \rightarrow K_2$. For $x \leq \beta$ put $U_x = u\{z: z < x\}$, $V_x = v\{z: z < x\}$. If $x, y \leq \beta$, let $G(x, y)$ consist of all $g: [M, P_2]$ such that $A_1 \subset M \subset S$, $g \upharpoonright A_1 = f$, $\text{dom } g \setminus A_1 = U_x \cup g^{-1}(V_y)$. Put $G = \bigcup (G(x, y): x \leq \beta, y \leq \beta)$. It is easy to see that G , ordered by inclusion (cf. 1.1 A), contains a maximal element, say h . Put $x_0 = \sup \{x \leq \beta: h \in G(x, y) \text{ for some } y\}$, $y_0 = \sup \{y \leq \beta: h \in G(x, y) \text{ for some } x\}$. It is easy to show that $h \in G(x_0, y_0)$. - Suppose that $x_0 < \beta$. Then $u(x_0) \notin \text{dom } h$. Hence, due to $(S_2, S_2) \in \mathcal{E}(\alpha, 2)$ and $\text{card } (\text{dom } h) < \alpha$, there is an isometric mapping $h': (\text{dom } h \cup \{u(x_0)\}) \rightarrow S_2$ such that $h' \upharpoonright \text{dom } h = h$. We have $h' \in G(x_0 + 1, y_0) \subset G$. This contradicts the fact that h is maximal, and proves that $x_0 = \beta$. The proof of $y_0 = \beta$ is analogous. Hence, $\text{dom } h \supset K_1$, $\text{dom } h^{-1} \supset K_2$, which proves the lemma.

2.6. Proposition. Let $\alpha \geq \omega$. If $S \in \mathcal{E}(\alpha)$ is complete and $\omega S \leq \alpha$, then S is Urysohn universal.

Proof. By 2.3, S is α -universal, hence, being complete, strongly α -universal. Let $A_1 \subset S$, $A_2 \subset S$, let $f: A_1 \rightarrow A_2$ be an isometry and let $\text{card } A_1 < \alpha$. By 2.5, there are dense sets $T_1 \subset S$, $T_2 \subset S$ and an isometry $g: T_1 \rightarrow T_2$ such that $A_i \subset T_i$ and $g \upharpoonright A_1 = f$. Since T_i are dense and S is complete, there is an isometry $h: S \rightarrow S$ such that $h \upharpoonright \text{dom } g = g$, hence $h \upharpoonright A_1 = f$.

2.7. Lemma. If $\alpha > \omega$ and $S \in \mathcal{E}(\alpha)$, then S is complete.

Proof. Let $(x_n: n \in \mathbb{N})$ be a Cauchy sequence in S ; assume that $x_m \neq x_n$ if $m \neq n$. Let $M = \mathbb{N} \cup \{\omega\}$; put $\sigma(i, j) = \rho(x_i, x_j)$, $\sigma(\omega, i) = \sigma(i, \omega) = \lim_{n \rightarrow \infty} \rho(x_i, x_n)$. Then (M, σ) is a metric space. Due to $(S, S) \in \mathcal{E}(\alpha, 2)$, there exists a $g \in [M, S]$ such that $g(i) = x_i$ for $i \in \mathbb{N}$. Clearly, $x_n \rightarrow g(\omega)$.

2.8. Proposition. Let $\alpha > \omega$ be a regular cardinal and let $\alpha^\omega = \alpha$. Let $T \in \mathcal{M}$, $\text{card } T \leq 2^\omega$. Then $U_\alpha(T)$ is a complete Urysohn universal space of weight α .

Proof. Put $S = U_\alpha(T)$. By 2.2, $(S, S) \in \mathcal{E}(\alpha, \alpha^+)$, hence S is α -universal. By 2.7, S is complete. By 1.10, $\text{card } S \leq 2^\omega$, hence $\omega S \leq \alpha$. By 2.6, this proves the proposition.

2.9. Lemma. Let $\alpha \geq \omega$. If S is α -homogeneous α -universal, then $S \in \mathcal{E}(\alpha)$.

Proof. Let $M \in \mathcal{M}$, $A \subset M$, $M \setminus A = \{q\}$, $\text{card } A < \alpha$. Let $f \in [A, S]$. Since S is α -universal, there exists a $g \in [M, S]$. For $p \in g(A)$ put $h(p) = f(g^{-1}p)$. Then $h: g(A) \rightarrow f(A)$ is an isometry. Since S is α -homogeneous, there exists an isometry $v: S \rightarrow S$ such that $v \upharpoonright g(A) = h$. Clearly, $v \circ g \in [M, S]$, $(v \circ g) \upharpoonright A = f$. We have shown that $(S, S) \in \mathcal{E}(\alpha, 2)$.

2.10. Proposition. Let $\alpha \geq \omega$. Let both S_1 and S_2 be complete Urysohn universal spaces of weight α . Then $S_1 \cong S_2$.

Proof. Since $wS_1 = \alpha$, there exist, by 2.5, dense sets $T_1 \subset S_1$ and an isometry $g: T_1 \rightarrow T_2$. Since S_1 are complete, there is an isometry $h: S_1 \rightarrow S_2$.

2.11. Lemma. If $\alpha \geq \omega$ and $(S, \rho) \in \mathcal{E}(\alpha)$, then $wS \geq \alpha^\omega$.

Proof. I. Let $\beta < \alpha$. Let A be a set of cardinality α . Put $T = \cup \{B^x: x < \beta\}$. If $\xi \in A^x$, $\eta \in A^y$, $x < y \leq \beta$, and $\eta \upharpoonright x = \xi$, put $\xi < \eta$. Let G consist of all $g: M \rightarrow S$ such that (1) $M \subset T$, (2) if $\xi < \eta$, $\eta \in M$, then $\xi \in M$, $\rho(g\xi, g\eta) = 1$, (3) if $x < \beta$, $\xi \in A^{x+1} \cap M$, $\eta \in A^{x+1}$, $\eta \upharpoonright x = \xi \upharpoonright x$, $\eta \neq \xi$, then $\eta \in M$, $\rho(g\xi, g\eta) = 2$.

II. It is easy to see that G contains a maximal element, say h . We will suppose that $\text{dom } h \neq T$ and derive a contradiction. Let x be the least ordinal such that $A^x \setminus \text{dom } h \neq \emptyset$. Choose a $\xi \in A^x \setminus \text{dom } h$. Clearly, each $\eta < \xi$ is in $\text{dom } h$. - If x is a limit ordinal, then, due to $(S, S) \in \mathcal{E}(\alpha, 2)$, there is a point $q \in S$ such that $\rho(q, h\eta) = 1$ for all $\eta < \xi$. Put $h'(\xi) = h(\xi)$ if $\xi \in \text{dom } h$, and put $h(\xi) = q$. It is easy to see that $h' \in G$, which contradicts the maximality of h . - Consider the case $x = z + 1$. Since, by 2.3, $(S, S) \in \mathcal{E}(\alpha, \alpha^+)$, there exists a set $K \subset S$ of cardinality α such that $\rho(p, q) = 2$ whenever $p, q \in K$, $p \neq q$, and $\rho(p, h\eta) = 1$ whenever $\eta < \xi$. Let $f: A \rightarrow K$ be a bijection; put $h'(\xi) = h(\xi)$ if $\xi \in \text{dom } h$, and $h'(\xi) = f(\xi(z))$ whenever $\xi \in A^x$, $\xi \upharpoonright z = \xi \upharpoonright z$. It is easy to prove that $h' \in G$, which contradicts the fact that h is maximal. Thus, we have shown that $\text{dom } h = T$.

III. If $\xi \in A^\beta$, choose a point $p(\xi) \in S$ such that $\rho(p(\xi), h\eta) = 1/2$ for all $\eta < \xi$ (this is possible, due to $(S, S) \in \mathcal{E}(\alpha, 2)$). Put $P = \{p(\xi): \xi \in A^\beta\}$. If $\xi, \zeta \in A^\beta$, $\xi \neq \zeta$, let x be the least ordinal such that $\xi(x) \neq \zeta(x)$. Put $z = x + 1$,

$\xi' = \xi \upharpoonright z$, $\zeta' = \zeta \upharpoonright z$. By (3), we have $\rho(h\xi', h\zeta') = 2$. Since $\xi' < \xi$, $\zeta' < \zeta$, we have $\rho(p(\xi'), h\xi') = 1/2$, $\rho(p(\zeta'), h\zeta') = 1/2$, and therefore $\rho(p(\xi), p(\zeta)) \geq 1$. This proves that P is a discrete subspace of cardinality α^β . Hence, $\beta < \alpha$ being arbitrary, we have $wS \geq \alpha^\omega$.

2.12. Proposition. Let $\alpha \geq \omega$. If S is an α -homogeneous α -universal space, then $wS \geq \alpha^\omega$.

This follows at once from 2.9 and 2.11.

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3.1. Fact. Let $T \in \mathcal{M}$ and let $\varphi \in \text{Is}(T, T)$. Then there is an isometry Φ of $U_\omega(T)$ such that $\Phi \upharpoonright T = \varphi$ and all $U_n(T)$ and $E(U_n(T), m)$, where $n \in \mathbb{N}$, $m > 1$, are invariant under Φ .

Proof. Put $T_n = U_n(T)$. Define φ_n as follows: $\varphi_0 = \varphi$; if $n \in \mathbb{N}$, then $\varphi_{n+1} = E(\varphi_n) \upharpoonright T_{n+1}$. For $p \in U_\omega(T)$, let $\Phi(p) = \varphi_n(p)$, where n satisfies $p \in T_n$. It is easy to see that we obtain a mapping Φ which has the properties stated above.

3.2. Proposition. Let $T \in \mathcal{M}$, $wT \leq \omega$. Let $T_0 = T$; for $n \in \mathbb{N}$, let T_{n+1} be a completion of $U_\omega(T_n)$. Put $P = \cup T_n$. Then (1) P is Urysohn universal, $wP = \omega$, (2) if $n \in \mathbb{N}$ and $\varphi \in \text{Is}(T_n, T_n)$, then $\Phi \upharpoonright T_n = \varphi$ for some isometry Φ of P .

Proof. I. Put $S_n = U_\omega(T_n)$. From 1.8, it follows, by induction, that $wT_n = \omega$ for all $n \in \mathbb{N}$, hence $wP = \omega$. For each $n \in \mathbb{N}$, by 2.2, $S_n \in \mathcal{E}(\omega)$, hence S_n is ω -universal. This implies that each T_{n+1} , hence also P , is strongly ω -universal. - II. We are going to prove the assertion (2). Put $\varphi_n = \varphi$. For $k \geq n$, put $\varphi_{k+1} = E(\varphi_k) \upharpoonright S_{k+1}$ and let φ_{k+1} be the isometry of T_{k+1} such that $\varphi_{k+1} \upharpoonright S_k = \varphi_k$. In this way, we get isometries $\varphi_k \in \text{Is}(T_k, T_k)$, where $k \geq n$, such that $\varphi_{k+1} \upharpoonright T_k = \varphi_k$. For any $p \in P$, put $\Phi(p) = \varphi_m(p)$, where $m \geq n$, $p \in T_m$. Clearly, Φ is an isometry of P , $\Phi \upharpoonright T_n = \varphi$. - III. It remains to show that P is ω -homogeneous. Let $A \subset P$, $B \subset P$, $0 < \text{card } A < \omega$, and let $f \in \text{Is}(A, B)$. Choose $n \in \mathbb{N}$ such that $A \cup B \subset S_n$. Since $S_n \in \mathcal{E}(\omega)$, there are, by 2.5, dense subsets K and M of S_n and $g \in \text{Is}(K, M)$ such that $g \upharpoonright A = f$. Hence, there is an isometry φ of T_{n+1} such that $\varphi \upharpoonright A = f$. By II, there is an isometry Φ of P such that $\Phi \upharpoonright T_{n+1} = \varphi$.

3.3. Lemma. If S is a complete space, $wS = \omega$, T is dense in S and $T \in \mathcal{E}(\omega)$, then $S \in \mathcal{E}(\omega)$.

We omit the proof, since its main idea is that of P. Ury-

sohn's proof of Theorem I in [3].

3.4. Proposition. Let S be a complete space, let $wS = \omega$ and let TcS be dense in S . If T is Urysohn universal, then so is S .

Proof. By 2.9, $T \in \mathcal{E}(\omega)$. By 3.3, $S \in \mathcal{E}(\omega)$. By 2.6, S is Urysohn universal.

3.5. Theorem I is an immediate consequence of 3.2 and 3.4.

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The category of ordered Cauchy spaces is cartesian closed

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Cauchy structures [6] provide a useful tool in the theory of lattice completions (see for instance [1]). In part therefore, the authors were led to study natural compatibilities between an order relation and a Cauchy structure on the same set (cf. [7], [8], and [9]).

This note will establish the cartesian closedness property for two ordered Cauchy space categories, here denoted by $PCHY$ and $OCHY$, and their related convergence space categories. Quite lately, the category of Cauchy spaces was proved to be cartesian closed (see [2]).

For cartesian closedness and applications, the survey paper [4] is recommended.

1. Preliminaries

1.1. The categories $PCHY$ and $OCHY$

Let X be a set.

For $\mathcal{F}$ and $\mathcal{G}$ filters (i.e. set theoretical proper filters) on X , the *g.l.b. filter* is denoted by $\mathcal{F} \cap \mathcal{G}$, and the *l.u.b. filter* (if it exists, i.e. if it is a proper filter) by $\mathcal{F} \cup \mathcal{G}$. For $x \in X$, $[x]$ denotes the trivial ultrafilter generated by $\{x\}$.

The pair $(X, \mathcal{C})$ is a *Cauchy space* [6], if $\mathcal{C}$ is a family of filters on X satisfying

- (c1) $[x] \in \mathcal{C}$ for all $x \in X$
- (c2) $\mathcal{F} \in \mathcal{C}$ and $\mathcal{G} \supseteq \mathcal{F} \Rightarrow \mathcal{G} \in \mathcal{C}$
- (c3) $\mathcal{F}, \mathcal{G} \in \mathcal{C}$ and $\mathcal{F} \cup \mathcal{G}$ exists $\Rightarrow \mathcal{F} \cap \mathcal{G} \in \mathcal{C}$.

The set $\mathcal{O} \subseteq X \times X$ is a *pre-order*, if it is reflexive and transitive, and an *order*, if it is anti-symmetric also. For $A \subseteq X$, $\mathcal{O}(A)$, $\mathcal{O}^{-1}(A)$, and $\mathcal{O}(A) \cap \mathcal{O}^{-1}(A)$ are the increasing, decreasing, and convex hull of A , respectively.

Passing to order compatibility, let $\mathcal{C}$ be a Cauchy structure and $\mathcal{O}$ a pre-order on X . For $\mathcal{F}, \mathcal{G} \in \mathcal{C}$, denote $\mathcal{F} \lesssim \mathcal{G}$, if there is a finite sequence $\mathcal{H}_1, \mathcal{H}_2, \dots, \mathcal{H}_n$ of filters in $\mathcal{C}$ such that $\mathcal{O}(\mathcal{F}) \cup \mathcal{H}_1, \mathcal{O}(\mathcal{H}_1) \cup \mathcal{H}_2, \dots, \mathcal{O}(\mathcal{H}_n) \cup \mathcal{G}$ all exist.

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